

Melanoma Decision Support System (CDSS)

Video Demonstration Script & Grad-CAM Technical Guide · Group 15 (E/22)

Part 1: What Does Grad-CAM Show in the Web Application?

When professors or clinicians ask what the Grad-CAM visualization represents, use this technical and clinical summary:

Visual Region	Gradient Meaning	Clinical Interpretation (ABCD Rule Correlation)
Red / Orange (Hot Spots)	High positive gradient activation flowing into layer 4 conv2.	Visual patterns that strongly convinced the network to predict Malignant Melanoma : asymmetric borders, atypical pigment networks, and focal dark clusters.
Cyan / Blue (Cold Regions)	Zero or negligible gradient activation.	Uniform, regular areas or surrounding healthy skin that did not contribute to malignancy suspicion.

Why This Matters Clinically: Deep learning models are traditionally perceived as opaque "black boxes". Grad-CAM provides visual verification that the network is making decisions based on genuine dermatological pathology rather than irrelevant background artifacts like skin hair, ruler marks, or dermatoscope gel bubbles.

Part 2: Scene-by-Scene Spoken Demonstration Script (Humanized)

This script is formulated in a **natural, confident conversational tone**. Follow the on-screen action prompts while speaking:

Scene 1: Introduction & Clinical Motivation (0:00 – 0:30)

[On Screen]: Show main dashboard at <http://127.0.0.1:5000>. Circle mouse around title and the green status pill.

[What to Say]: "Hi everyone. Today I'm demonstrating our melanoma detection web app, which we built for our CO5430 Computer Vision project at the University of Peradeniya.

Skin cancer detection is all about catching it early. If melanoma is caught in time, survival is over 98%. But if a doctor misses it, the consequences can be fatal. That's why we didn't just build another image classifier—we built a genuine clinical tool focused on **not missing any cancer cases**, while making sure the AI can explain why it made its decision.

Up here in the corner, you can see our fine-tuned **ResNet-18 PyTorch model** is loaded and running live on the backend."

Scene 2: Batch Upload & Patient Triage Queue (0:30 – 1:05)

[On Screen]: Click "Load 10 Reference Cases". Point to top KPI cards updating, then scroll down through the table.

[What to Say]: "In a real clinic, doctors don't just test one image at a time. So we designed this as a batch workstation. I can drag and drop multiple skin images at once, or load our dataset reference cases right here.

As soon as they load, each image gets analyzed in a fraction of a second.

The top cards give us a quick overview—how many total cases we have, how many came back suspicious, and how many look benign. Down in the queue, every patient case gets a clear verdict, a malignancy probability score bar, and an immediate clinical recommendation—like whether the patient needs an urgent biopsy or just a routine check-up."

Scene 3: Threshold Calibration Switch (1:05 – 1:40)

[On Screen]: Click between $\tau = 0.35$ (Clinical Safety) and $\tau = 0.50$ (Standard) at top right. Observe table badges updating.

[What to Say]: "Now, this button right here is one of the most important parts of our project.

Normally, machine learning models use a 50% cutoff—meaning anything over 0.5 is called cancer, and anything under is called safe. But in medicine, being 50/50 isn't good enough when a missed melanoma is life-threatening.

So during our research, we calibrated the threshold down to **0.35**. By being slightly more cautious, our model's sensitivity jumped to **97.2%**, cutting down missed melanomas by **84%**.

And notice that when I switch between Standard mode and Safety mode, the system updates the entire patient queue instantly right in front of us, without having to re-upload anything."

Scene 4: Explainable AI & Grad-CAM Inspection (1:40 – 2:30)

[On Screen]: Click "Inspect Grad-CAM" on a Malignant row. Adjust opacity slider, switch tabs, point to ABCD metrics.

[What to Say]: "A big problem with deep learning in healthcare is that doctors don't trust 'black boxes'—they need to know why the AI flagged an image. That's where **Grad-CAM** comes in.

If I click 'Inspect Grad-CAM' on this case, it opens up our visual analysis modal. What you're seeing here isn't just a random colored filter—it's the actual gradient map from the deepest convolutional layer of our ResNet-18 model.

The bright red and yellow hotspots show exactly what the network focused on. Notice how the heat concentrates right around the jagged, asymmetric edges and the dark clusters of pigment. That tells the doctor the AI is looking at real pathological warning signs, not just background skin or lighting artifacts.

We can adjust the opacity slider to see the raw image underneath, or switch to the pure heatmap view. And on the right side, we broke down the classic dermatological **ABCD rule**—scoring Asymmetry, Border, Color, and Diameter."

Scene 5: Export & Conclusion (2:30 – 2:55)

[On Screen]: Close modal. Click "Export Clinical Audit (CSV)", then "Print Case Summary" showing print preview.

[What to Say]: "Finally, to make this practical for clinical record-keeping, the doctor can click **Export Clinical Audit** to download a clean CSV file with all the patient scores and threshold settings, or hit **Print Case Summary** to get a clean printable report for the patient's medical file.

Overall, we took our trained deep learning model and turned it into an interactive, clinically safe decision-support system that dermatologists can actually understand and trust.

Thank you so much for your time!"

Part 3: Practical Presentation Tips for Viva & Recording

- **Pacing:** Speak in a calm, conversational rhythm. Pause for 1 second after clicking a button so the viewer's eye can follow your actions.
- **Mouse Guidance:** Use your cursor as a pointer. Hover intentionally over the threshold button, the MSI progress bar, and the Grad-CAM hotspots while speaking about them.
- **Key Phrase to Remember:** "We calibrated our threshold to 0.35 because in clinical oncology, sensitivity and catching every melanoma matters more than raw accuracy."
- **Visual Verification:** Ensure the application is maximized in your browser at 1080p resolution for maximum clarity.